

AVL

Insert 9: → Insert 2: → Insert 1:

⑨
Balanced: First Node

② ⑨
Balanced

① ② ⑨
Unbalanced → Single Right Rotation at 2 → ② ① ⑨

Insert 10: → Insert 8: → Insert 3:

① ② ④ ⑩
Balanced No Rotation Needed

① ② ④ ⑧ ⑩
Balanced No Rotation Needed

① ② ④ ⑧ ⑩ ③
Double Rotation: Right Rotation at 4, Left Rotation at 2

Rotation 1 → Rotation 2

① ② ④ ⑧ ⑩ ③
① ② ④ ⑧ ⑩ ③

Insert 7:
① ② ④ ⑧ ⑩ ③ ⑦
Balanced. No Rotation Needed

→ Insert 4: → Rotation 1 - Right Rotate 7 → Rotation 2: Left Rotate 3

① ② ④ ⑧ ⑩ ③ ⑦ ④
Before Balancing Node 3 is unbalanced

① ② ④ ⑧ ⑩ ③ ⑦ ④

Insert 4 causes a right left imbalance at Node 3

~~① ② ④ ⑧ ⑩ ③ ⑦ ④~~ ① ② ④ ⑧ ⑩ ③ ⑦ ④

→ Insert 6: → Single Left Rotation at 2: → Insert 5:

① ② ④ ⑧ ⑩ ③ ⑦ ④ ⑥
unbalanced at 2

① ② ④ ⑧ ⑩ ③ ⑦ ④ ⑥

Insert 6 makes node 2 right-heavy. use single left rotation at 2

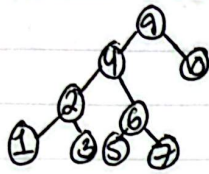
① ② ④ ⑧ ⑩ ③ ⑦ ④ ⑥ ⑤
Node 7 too left-heavy

→ Right Rotation at 7

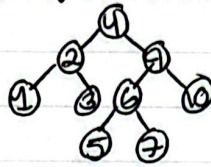
① ② ④ ⑧ ⑩ ③ ⑦ ④ ⑥ ⑤

AVL

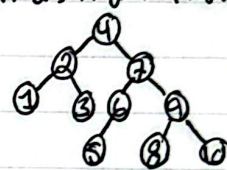
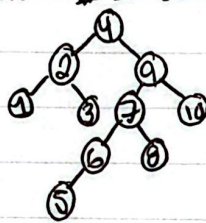
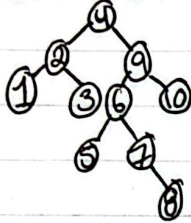
→ Remove Root(8): → Single Right Rotation at 9:



Very heavy imbalance, 9 is root

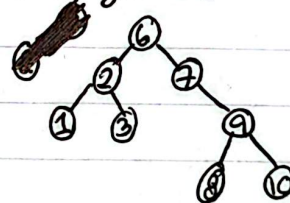
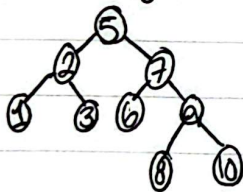


→ Insert 8 Back: → Rotation 1: Left Rotate at 6: → Rotation 2: Right Rotate at 9:



Reinsert 8. Node 9 becomes left-right Heavy. First perform a left rotation at 6 then a right rotation at 4

Delete Root Again (4): → Delete Root Again (5):



Delete root 4, inorder Successor is 5. AVL is Balanced

→ Left Rotate at 7:

